

You will act as a school teacher who understands the teaching goals of a middle school Physics class.

Here are two simulations that the students are working on. Please go over them thoroughly: ‘Projectile’ ([Projectile Motion](#)) and ‘Coulumb’s Law’ ([Coulomb's Law](#)). Students will be asked to answer questions based on this experiment. To better answer those, the students will do these sets of activities.

From an Excel file, you will receive:

- **Variables** for these 2 simulations.
- **Relationships** among these variables.
- **Questions** for these 2 simulations that the students need to answer after performing the activities.
- **Activities** (2 sets each) for these 2 simulations.

Your task is to score each set of activities across five evaluation criteria. Assign a score out of 10 along with a justification of 1-2 sentences for activity sets (labelled ‘Level 2’ and ‘Level 3’) for each of the 2 simulations (rows) for each of the following 5 evaluation criteria:

1. Concept & Variable Coverage: How well do the activities reflect and incorporate the listed variables and relationships?
2. Semantic Relevance to Questions: How well do the activities help students answer the provided questions?
3. Conceptual Progression: Does the activity set have a clear logical order where the sequential activity builds upon the previous activity?
4. Clarity & Design Precision: Does the activity set give a clear instruction (e.g., “Vary X while keeping Y constant”, “measure Z and record”) and avoids vague phrases like “try something” or “explore freely”?
5. Output Format Accuracy: Are the activities clearly separated, numbered or bulleted?

Sample response:

| ****Concept & Variable Coverage**** | 7/10 | The activities effectively cover the core input variables (Initial Velocity, Launch Angle) and their relationship to key outputs (Range, Time of Flight). However, they miss opportunities to explore gravity or air resistance. |

| ****Semantic Relevance to Questions**** | 8/10 | The activities are well-aligned with the questions, directly guiding students to explore the optimal launch angle and the effects of changing initial velocity on range. |

| ****Conceptual Progression**** | 6/10 | The set has a decent progression from exploring angles to exploring initial velocity. However, the flow could be more integrated, and the final activity is incomplete. |

| ****Clarity & Design Precision**** | 6/10 | Instructions are functional but lack precision (e.g., "Launch projectiles at various angles" without specifying which ones). The inclusion of controlled variables is a plus but isn't consistently applied. |

| ****Output Format Accuracy**** | 8/10 | The activities are clearly numbered and separated with a consistent structure. The score is reduced because the final activity is cut off mid-sentence. |